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Measuring University Patent Technological Innovation Capability from Patent Knowledge Association Structures

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Abstract

Introduction. Evaluation of university patent technological innovation capability has long relied on result-oriented indicators such as patent counts, grants, and citation frequency. These indicators reflect mainly the scale or external impact of innovative output and reveal little about the internal capabilities—knowledge sourcing, recombination, and diffusion—that operate during the innovation process. As university patent output keeps growing, identifying this capability and its formation mechanisms from the knowledge-association structures carried by patents has become a central problem in innovation evaluation. Conventional approaches under-represent the formation process, limiting understanding of how innovation advantages arise and how resources should be allocated differentially.

Method. This paper proposes a measurement framework for university patent technological innovation capability based on patent knowledge-association structures. Unlike evaluation built on a single output indicator or a single patent relation, patents are treated as relational objects embedded in a technical-knowledge system. Three relation layers—IPC classification, semantic similarity, and citation linkage—are fused into a multi-dimensional knowledge-association structure, and capability is characterized along five dimensions: knowledge sourcing, knowledge recombination, technological exploration, knowledge diffusion, and dynamic transformation.

Analysis. A multi-relation patent knowledge-association structure is constructed from invention-patent data for 124 Chinese Double-First-Class universities over 1990–2019. Capability profiles are computed

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per university-period, their structural features are described, and development patterns are identified through a priori criteria cross-checked with Gaussian-mixture grouping and first-order Markov transitions, controlling for patent volume, technological composition, and coverage.

Results. Universities with similar patent output differ significantly in their knowledge-capability structures, showing that traditional output indicators do not fully reveal university patent technological innovation capability. Three typical development patterns are identified—steady utilization, exploratory innovation, and structural recombination—revealing distinct innovation paths in knowledge accumulation, cross-domain recombination, and technological diffusion.

Conclusion. The contribution is to extend university patent innovation-capability measurement from static output analysis to process-based analysis grounded in knowledge-association structures, offering a new perspective on the formation mechanisms of university innovation capability and methodological support for innovation evaluation and the optimal allocation of scientific and technological resources.

Introduction

Universities with the same number of patents can produce them through different knowledge processes: one draws on a narrow set of external sources and refines an established core; the other connects technical elements that rarely appear together and is later diffused across organizations outside its own field. Patent counts, grants, and citation frequency record the output of both institutions but not the difference between them (Griliches, 1990; Hall et al., 2005; Nagaoka et al., 2010). Evaluation of university patent technological innovation capability has therefore long relied on result-oriented indicators that reflect the scale or external impact of output, while leaving the internal capabilities that operate during the innovation process—knowledge sourcing, recombination, and diffusion—largely unobserved. As university patent output keeps growing, identifying this capability and its formation mechanisms from the knowledge-association structures carried by patents has become a central problem in innovation evaluation, and one that conventional approaches, which under-represent the formation process, leave unresolved.

Information science already treats documents as relational objects. Bibliographic coupling and co-citation identify intellectual connections (Kessler, 1963; Price, 1965; Small, 1973); knowledge maps represent domains and their change over time (Börner et al., 2003; Chen, 2006; van Eck & Waltman, 2010); patent overlays extend these methods to technology (Kay et al., 2014; Leydesdorff et al., 2014; Rafols et al., 2010; Yan & Luo, 2017). What is missing is a way to move from a map, which shows where an institution is located at one moment, to a capability measure that shows how its knowledge relations are sourced, combined, explored, diffused, and

reorganized—and whether observed change exceeds sampling and measurement noise.

This paper proposes such a measurement framework for Chinese university patents. The central idea is simple. Patent relations of different kinds—shared classifications, textual proximity, and citations—express different knowledge relationships and become observable at different times. Fusing them into a multi-dimensional knowledge-association structure, and dating each one, makes it possible to characterize a university's patent technological innovation capability along five dimensions: *knowledge sourcing* (the reach and diversity of external knowledge a university builds on), *knowledge recombination* (the rarity of the technical-element combinations it forms), *technological exploration* (the novelty of the directions it enters relative to its own past), *knowledge diffusion* (the breadth and uptake of its knowledge among later users), and *dynamic transformation* (sustained reorganization of its technological composition across periods). The first four are observable when an invention is published or shortly after; the fifth is assessed independently across periods against a within-university null, so that future structural change is not used to construct a state that is then said to explain that change. The two research questions are:

RQ1. How can patent knowledge-association structures be used to measure different dimensions of university patent technological innovation capability?

RQ2. To what extent can a knowledge-association-based framework reveal university patent technological innovation capability beyond conventional output-oriented indicators?

Using a longitudinal corpus of invention patents from 124 Chinese Double-First-Class universities (1990–2019), this paper demonstrates the empirical validity of the framework. The core contribution is a traceable link between knowledge organization and capability measurement: every profile value and pattern statement can be followed back to dated patent records and the specific relations that support it. The paper specifies the framework, the measures, and the reporting design, while explicitly identifying what a university-only patent and citation corpus can and cannot support.

Related work

Patents are selective records of knowledge production. Not all inventions are patented, quality varies, and legal and organizational strategies shape the record (Griliches, 1990; Nagaoka et al., 2010). Citations require particular care: examiner-added references weaken their reading as inventor learning (Alcácer & Gittelman, 2006), and knowledge-flow estimates depend on technological controls and comparison groups (Jaffe et al., 1993; Thompson & Fox-Kean, 2005). Patents are therefore treated here as documented technical objects with dated relations, not as complete histories of invention.

Innovation-capability evaluation has nonetheless relied heavily on these same output indicators. Patent counts, grant counts, and citation frequency summarize the

scale or external impact of inventive output, but universities producing the same volume can differ in how they source, combine, and diffuse technical knowledge (Griliches, 1990; Hall et al., 2005). Aggregating across heterogeneous knowledge processes into a single performance score obscures these differences and offers little leverage on how innovation advantages form. The framework here keeps the underlying relations visible so that capability is read from structure rather than summarized by count.

Three relation types are used in patent mapping. Classification systems organize technical categories (Leydesdorff, 2008); text exposes similarities that shared codes miss (Kelly et al., 2021; Sarica et al., 2020; Whalen et al., 2020); citations record sourcing and later use. Overlay maps place organizations in a common knowledge space (Kay et al., 2014; Rafols et al., 2010), and relatedness research links positions in that space to entry and exit (Kogler et al., 2013; Rigby, 2015). Layout movement, however, can arise from a changed sample or algorithm rather than from changed knowledge. This study keeps visual coordinates fixed and measures change in the underlying distributions.

The process vocabulary comes from innovation research. Absorptive capacity concerns the recognition and use of external knowledge (W. M. Cohen & Levinthal, 1990); exploration and exploitation (March, 1991) and search depth and scope (Katila & Ahuja, 2002) describe learning orientations; combinatorial and atypical-combination studies motivate measuring unusual connections separately from their later impact (Fleming, 2001; Uzzi et al., 2013; Verhoeven et al., 2016; Wang et al., 2017; Youn et al., 2015). Emergence is more than novelty: its attributes include growth, coherence, impact, and uncertainty (Rotolo et al., 2015), of which scientometric methods operationalize a subset (Small et al., 2014). First citation and later accumulation carry different temporal information (Huang et al., 2019). Multilayer network research shows why heterogeneous relations should not be summed into one adjacency matrix (Boccaletti et al., 2014; Kivela et al., 2014). The present framework draws on these sources but reframes two constructs for a capability-structure reading: the novelty-growth signal is read as *technological exploration* of new directions rather than field emergence, and the breadth and uptake of later citations is read as *knowledge diffusion* rather than bounded reuse. Throughout, patent references index documented sourcing, not capability; rare code pairs index unusual combination, not usefulness; later citing records index bounded diffusion, not learning.

Framework: relations, time, and transformation

University patent technological innovation capability is defined here as the capacity to source, combine, explore, diffuse, and reorganize technical knowledge, as observed in the relations carried by a university's patents. Patents are the observable objects; universities are the contexts in which they accumulate and connect. No patent or institution is assigned an intrinsic developmental type.

Figure 1 shows the design. Four concurrent dimensions describe a university-

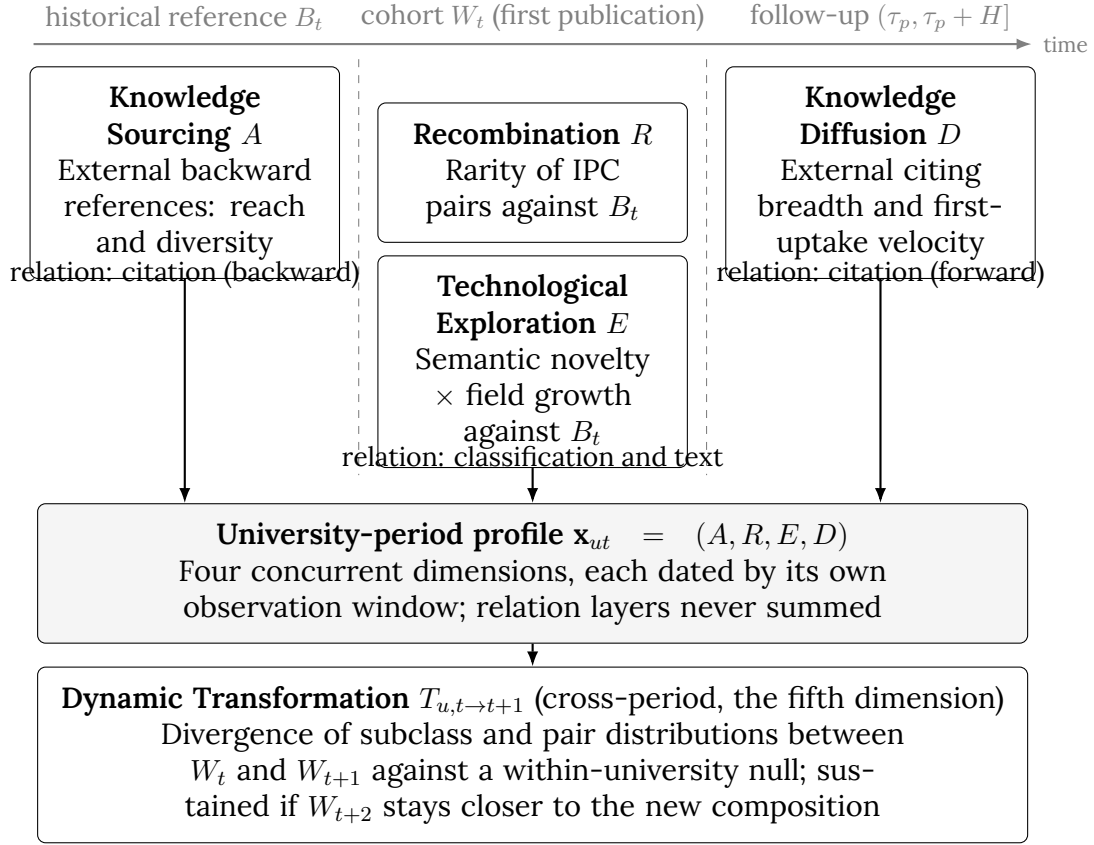


Figure 1. Framework. Four concurrent dimensions are placed at the time they become observable and are drawn from distinct patent relations; dynamic transformation is a separate cross-period statistic—the fifth dimension—kept outside the concurrent profile to avoid circularity. No arrow denotes a causal effect or a compulsory stage.

period profile,

$$\mathbf{x}_{ut} = (A_{ut}, R_{ut}, E_{ut}, D_{ut})^T, \quad (1)$$

where A (knowledge sourcing) and R (recombination) are observable when inventions are published, E (technological exploration) uses the current cohort against its own historical reference, and D (knowledge diffusion) requires a fixed follow-up horizon. The fifth dimension, dynamic transformation $T_{u,t \rightarrow t+1}$, is a cross-period statistic on the distribution of technical categories and pairs, assessed separately with its own null. Keeping T outside the concurrent profile prevents a circular reading in which future structural change is used to describe a state and the state is then said to explain that change; the five dimensions together still characterize the capability, four synchronously and one longitudinally.

Two temporal rules follow. First, each statistic is dated by the latest information required to determine it, not by the earliest observable signal: A and R are available when an invention is published, E becomes available only when its cohort closes (it uses the cohort's subclass-growth shares), and D requires a fixed five-year follow-up so that its cohort-relative ranks can mature. The profile $\mathbf{x}_{ut}$ is therefore a retrospective portrait of a closed cohort, not a contemporaneously observable state at the cohort date, and no indicator uses information published after its own observation window. Second, a profile at one period is a state, not a

Coverage item	Value	Required definition or audit
Database/provider; extraction date	TBD	Frozen source and query manifest
Retrieval dates; analytic dates	TBD	First-publication cohorts; mature end date
Double-First-Class roster (124); linked institutions	TBD	Dated roster and name crosswalk
Raw publication records	TBD	Before publication/application deduplication
Eligible invention units	TBD	Families, or applications if families unavailable
Eligible university-periods	TBD	Three-year non-overlapping cohorts
IPC subclasses; main groups	TBD	Version-harmonized vocabulary sizes
Usable title/abstract records	TBD	Count and coverage percentage
Backward-citation links	TBD	Resolved earlier public invention units
Forward-citation links within H	TBD	Unique family pairs, scope declared
External citing organizations	TBD	University-only or broader coverage
Excluded units by reason	TBD	Mutually exclusive first-failure counts

Table 1. Dataset scale and coverage, to be computed from the frozen source database. No value or date range in this table is an empirical estimate.

trajectory; trajectory statements require ordered profiles for the same institution and evidence that observed change exceeds a within-university null.

Three interpretive development patterns are retained as *hypotheses to be checked against these observables*, not as classes to be imposed. A *steady-utilization* pattern shows a stable technical core: low T , low E , and moderate-to-high D on an established knowledge base. An *exploratory-innovation* pattern shows high E and/or high R , that is, the university repeatedly enters novel technological directions and forms unusual combinations. A *structural-recombination* pattern requires T above its null in at least one transition and persistence of the new composition in the following period, indicating a sustained reorganization of the portfolio. An institution may follow different patterns at different times, mixed cases remain visible, and the patterns form no performance ranking.

Data and observable relations

Corpus and its boundaries

The source is the research team’s database of Chinese university invention patents and their citation records, covering 124 Double-First-Class universities over 1990–2019. Table 1 lists the coverage items that must be audited before analysis. Its rows determine which relations in the framework are observable and are therefore part of the design rather than a formality.

The primary cohort comprises published Chinese invention applications with at least one university applicant at first publication, restricted to the 124 Double-First-Class universities over 1990–2019, regardless of later grant status; utility models and designs are excluded. Institutional coverage is defined by a dated roster of the 124 Double-First-Class universities, with historical names, mergers, and affiliated hospitals resolved through time-specific linkage; ambiguous matches remain auditable rather than silently assigned. Publication and grant versions of an application are collapsed. If family identifiers are available, the invention unit

is a simple family represented by its earliest eligible document; otherwise it is the deduplicated application, and the unit is declared. A patent with m_p university applicants contributes $\omega_{up} = 1/m_p$ to each; full counts are a sensitivity.

Three properties of a university-only corpus constrain the design and are stated here rather than discovered later. (i) Backward references from university patents are observable, but the cited documents' own attributes are available only when the cited record is itself in the corpus or its classification is carried in the reference field. (ii) Forward citations to university patents are observable; the citing documents' applicants are observable only where the citation record carries applicant fields, and the citing documents' own forward links are generally not available. Reuse depth beyond one citation step is therefore excluded from the primary measure. (iii) Chinese invention patents receive few forward citations in their first year, so any indicator conditioned on early recognition will be zero for most records. Emergence is consequently measured in an early-information form that does not depend on recognition, with a recognized variant as a secondary measure.

Observation time

First publication is the main clock. $W_t = [b_t, b_{t+1})$ is a non-overlapping three-year cohort and $B_t = [b_t - 3, b_t)$ its historical reference; reuse uses a horizon of $H = 5$ years. Mature profiles require $b_{t+1} + H \leq d_{\text{extract}}$, and cohorts without a full historical reference are excluded. The number of mature cohorts this leaves is a property of the database, to be reported in Table 1; if fewer than three consecutive mature cohorts exist, trajectory analysis is restricted to description and the transformation test is reported as infeasible rather than approximated.

A university-period enters profile analysis with at least 20 attributed inventions and at least 80% usable attributed mass for the fields each dimension requires. Confirmed absence of citations is zero; an unavailable field is missing and never enters a denominator. For dimension j the eligible subset is P_{ut}^j with mass $M_{ut}^j = \sum_{p \in P_{ut}^j} \omega_{up}$, reported as a share of total mass.

Three relation layers

Figure 2 shows how the corpus is decomposed. The technology layer has IPC main-group nodes linked by fractional co-occurrence,

$$w_{ij}^{(t)} = \sum_{p \in W_t: m_p^C \geq 2} \frac{\mathbb{I}\{i, j \in \mathcal{C}_p\}}{\binom{m_p^C}{2}}, \quad i < j, \quad (2)$$

where $\mathcal{C}_p$ is the set of distinct main groups on patent p ; each multi-coded invention contributes one unit of pair mass. The semantic layer has patent nodes linked by cosine similarity of character 2–4-gram TF-IDF vectors of normalized titles and abstracts, with vocabulary and IDF fitted on B_t only (features in at least five and at most 80% of reference documents, capped at 100,000; log TF, smoothed IDF, L_2 normalization). The citation layer has patent nodes with directed edges from

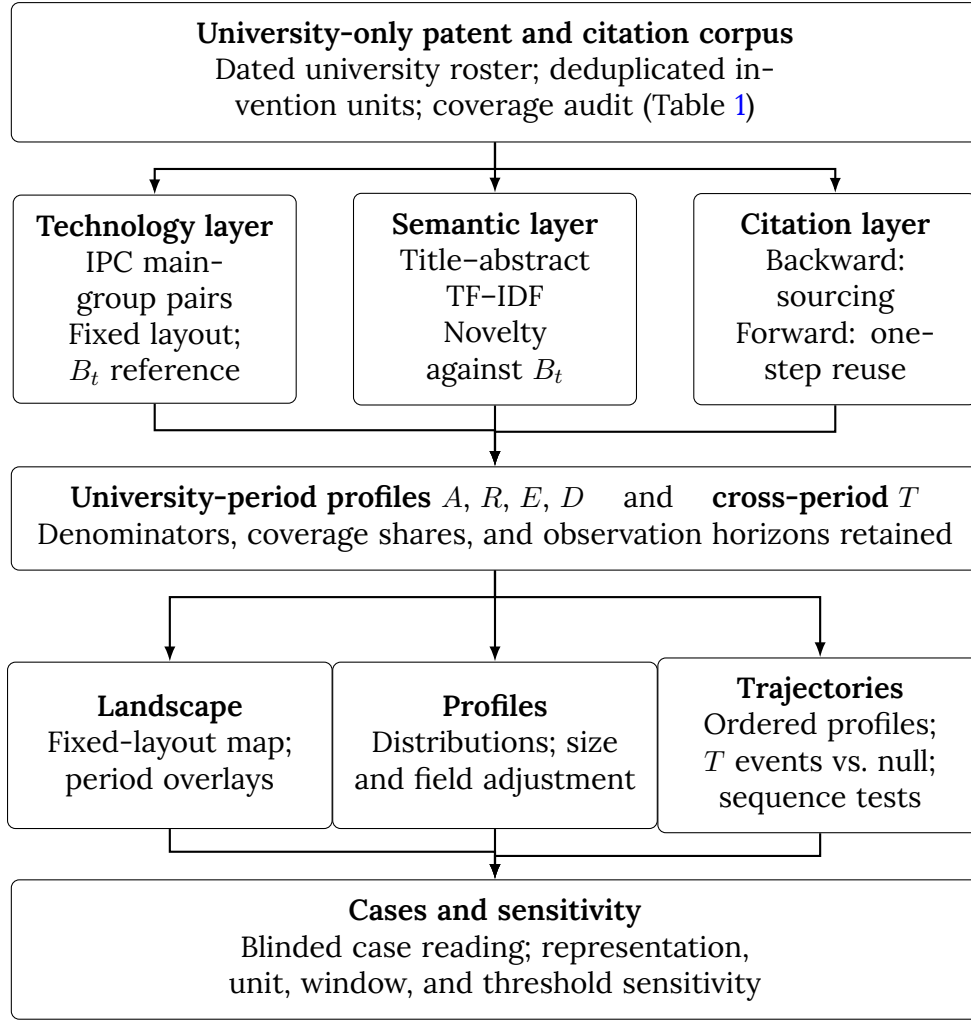


Figure 2. Data-to-inference workflow. Three relation layers are kept separate up to the profile; transformation is computed across periods from the technology layer alone.

cited to citing document, duplicate family pairs collapsed, within-family links removed, and citing publication required to be strictly later. Applicant-overlap self-citations are excluded from the primary reuse graph. A patent-IPC incidence relation connects the layers; their adjacency matrices are never summed. Display graphs (normalized edges, top-10 neighbor union; mutual 15-nearest-neighbor semantic graph; Leiden communities, Traag et al., 2019) use fixed coordinates across periods; all indicators use unpruned weights.

Measures

Knowledge sourcing

Let q_{ut} be the attributed share of valid inventions with at least one resolved external backward reference, and let $\pi_{ut,k}$ be the fractional distribution of those references

over a fixed, version-harmonized IPC subclass vocabulary of size K_t . Then

$$A_{ut} = q_{ut} \left[\frac{1}{2} + \frac{1}{2} \frac{-\sum_k \pi_{ut,k} \log \pi_{ut,k}}{\log K_t} \right], \quad (3)$$

so that confirmed absence of external sourcing gives $A = 0$ and the positive baseline distinguishes narrow sourcing from none. Both components are reported. A measures the reach and diversity of documented knowledge sourcing (Shannon, 1948; Stirling, 2007), not absorptive capacity (W. M. Cohen & Levinthal, 1990). The $\frac{1}{2}$ positive baseline is a reporting convention that keeps the no-sourcing and narrow-sourcing cases separable rather than a substantive equivalence between them; both components (q_{ut} and the diversity term) are reported so the weighting is auditable. When $K_t \leq 1$ or no classifiable external reference exists, the diversity term is undefined and the period is treated as missing rather than zero.

Recombination

Let $c_{ij,t}^-$ be the historical fractional pair mass of main groups (i, j) in B_t , Q_t the number of possible pairs in the pre-established vocabulary, and $C_t = \sum_{i < j} c_{ij,t}^- > 0$. With $\alpha = 1/2$,

$$p_{ij,t}^- = \frac{c_{ij,t}^- + \alpha}{C_t + \alpha Q_t}, \quad r_{ij,t} = \frac{-\log p_{ij,t}^-}{-\log\{\alpha/(C_t + \alpha Q_t)\}}, \quad (4)$$

$$R_{ut} = \frac{1}{M_{ut}^R} \sum_{p \in P_{ut}^R} \omega_{up} \frac{\sum_{(i,j) \in \mathcal{L}_p} r_{ij,t}}{|\mathcal{L}_p|}, \quad (5)$$

where $\mathcal{L}_p$ is the set of valid mapped pairs on patent p . Unresolved codes map to one recorded out-of-vocabulary category; a patent with no valid pair is missing, not zero. Rarity indexes unusual combination, not novelty or usefulness (Fleming, 2001; Verhoeven et al., 2016; Youn et al., 2015). Because a pair can be rare merely because its components are individually uncommon, R is interpreted conditional on a degree-preserving permutation null that holds marginal code frequencies; the observed-minus-permuted rarity is reported alongside R , and the share of single-coded patents (necessarily excluded from pairing) is reported as a coverage indicator. First-seen pair rates provide an additional check.

Technological exploration

Semantic novelty is $n_{p,t} = 1 - \max_{q \in B_t} s_{pq}$ over the full historical index, excluding same-family comparators. For subclass k , with fractional shares $\rho_{k,t}$ in the current cohort and $\rho_{k,t}^-$ in the reference,

$$g_{k,t} = \max \left\{ 0, \frac{\rho_{k,t} - \rho_{k,t}^-}{\rho_{k,t} + \rho_{k,t}^- + 10^{-12}} \right\}, \quad E_{ut} = \frac{1}{M_{ut}^E} \sum_{p \in P_{ut}^E} \omega_{up} n_{p,t} g_{p,t}, \quad (6)$$

where $g_{p,t}$ averages over the patent's subclasses. Both $n_{p,t}$ and $g_{k,t}$ are computed against the corpus-wide historical index B_t and current cohort W_t (field-level signals), not against a university's own past; "technological exploration" is therefore

read as the university producing work that is novel relative to the field’s history *and* located in growing subclasses, not as institutional unfamiliarity per se. Because $g_{p,t}$ uses the current cohort’s subclass shares, E becomes available only when the cohort closes; it is early-information in that it needs no citation or recognition, but it is retrospective at the cohort level rather than observable at a single patent’s publication. The novelty–growth product also weights exploration toward expanding directions, so novelty in stable or declining subclasses receives no growth credit; the recognized variant and case reading compensate. A recognized variant E^{rec} multiplies each product by an indicator of external citation within one year; given constraint (iii) in Section , it is reported with its nonzero share as a secondary measure and is not the primary exploration indicator. Neither variant covers the full construct of technological emergence (Rotolo et al., 2015; Small et al., 2014); technical coherence is assessed through cases.

Knowledge diffusion

Within $(\tau_p, \tau_p + H]$, let b_p count distinct external citing organizations where applicant fields are available and distinct external citing families otherwise (the level is recorded), and let $v_p = 1 - \Delta_p/H$ when the first external citation arrives after Δ_p years, with $v_p = 0$ for uncited patents. Each component is transformed to its midrank among positive values in the same primary subclass and cohort ($\phi(0) = 0$; back-off to IPC section and then cohort below 30 positives), and

$$D_{ut} = \frac{1}{2M_{ut}^D} \sum_{p \in P_{ut}^D} \omega_{up} [\phi^b(b_p) + \phi^v(v_p)]. \quad (7)$$

D measures *documented knowledge diffusion among observed citers*. Midranks are rescaled to $[0, 1]$ so the two components share a common scale; where applicant fields are missing, externality cannot be established and such records are flagged and excluded from b_p rather than silently classified as external or treated as confirmed zeros. Because the citing side of a university-only corpus is incomplete, D does not measure economy-wide diffusion, and a two-step depth component is not part of the primary measure; it is computed as a sensitivity only where descendants’ forward links exist. First uptake and accumulated breadth carry different information (Huang et al., 2019) and are also reported separately.

Dynamic transformation

Let $\mathbf{a}_{ut}$ and $\mathbf{r}_{ut}$ be the university’s normalized subclass and pair distributions on shared vocabularies. Then

$$T_{u,t \rightarrow t+1} = \frac{\text{JSD}(\mathbf{a}_{ut}, \mathbf{a}_{u,t+1}) + \text{JSD}(\mathbf{r}_{ut}, \mathbf{r}_{u,t+1})}{2 \log 2} \in [0, 1], \quad (8)$$

with JSD the Jensen–Shannon divergence (Lin, 1991). A transformation event requires T above the 95th percentile of a within-university null that pools the two adjacent cohorts and permutes their period labels while preserving cohort sizes,

confirmed under size-matched subsampling. A *sustained* event additionally requires the third period to be closer to the new composition than to the old and still distinguishable from the old. Mergers and IPC revision events are audited before any event is retained. The null exchanges patents between the two adjacent periods conditional on the pooled portfolio and cohort sizes; it therefore tests period-label exchangeability, not the magnitude of year-to-year variation under a temporally structured process, and its validity depends on the absence of secular trends, coverage change, or field-wide transitions during the window. Because pair changes partly follow subclass-composition changes, the two JSD terms are reported separately as well as summed, so the incremental contribution of the pair term is visible; multiple university-transition tests are controlled at the family-wise level. T measures institutional reorganization of a portfolio, not a technological paradigm shift (Dosi, 1982; Funk & Owen-Smith, 2017); a large T may reflect disruption or contraction as readily as adaptive change, and diversity is a distinct property (Porter & Rafols, 2009; Stirling, 2007).

Analysis and reporting design

The analysis proceeds in five steps, each answerable from the corpus described above.

Landscape. The technology layer is laid out once on the pooled corpus (Figure 3A); period overlays reuse the same coordinates (Figure 3B) so that entry, persistence, and exit of technical groups reflect the data rather than the layout. Coverage growth is evaluated against rarefied subclass coverage and portfolio size, since a larger sample alone makes more fields visible. Disagreement between the semantic and classification views marks cases for inspection.

Profiles. Distributions, zero shares, and denominators of A, R, E, D are reported for eligible university-periods (Table 2), together with pairwise correlations and their association with log patent volume, field composition, and period. Residual variation after these adjustments is the evidence that relational profiles add information beyond size and field. Profiles are then displayed on their original scales with whole-university bootstrap intervals (Efron, 1979), and universities are placed in profile space (Figure 4A).

Patterns. Development patterns are identified in two ways and then compared. First, the three a-priori patterns of Section (steady utilization, exploratory innovation, structural recombination) are applied through their stated observable criteria using thresholds prespecified before inspection, and the counts of supported, mixed, and unsupported cases are reported. Second, a Gaussian mixture on the concurrent profile (A, R, E, D), with the number of components selected by BIC and cluster-wise stability (Fraley & Raftery, 2002; Hennig, 2007), classifies university-period profiles into GMM states. Because the a-priori patterns are trajectory-level (multi-period) while GMM states are period-level, the cross-check first derives a trajectory label per university from its GMM-state sequence under the same prespecified rules, then compares the two labelings by adjusted Rand index on the common unit (universities) (Table 3); GMM adequacy

is assessed for bounded and zero-inflated variables, skewed masses, and repeated observations from the same university. For universities with consecutive eligible cohorts, a first-order Markov transition matrix among the derived trajectory labels reports how universities move between labels over time (Table 4), with classification uncertainty and the selective availability of consecutive cohorts reflected in the estimates. Ordered profiles are displayed alongside T and its null (Figure 4B). The proposition that sustained exploration or recombination precedes structural transformation is tested by comparing, within each university, the observed frequency of a “high- R/E period followed by a sustained T event” with its frequency under order permutations of that university’s own profile values, recomputing T events from the reordered sequence. The contrast is reported with a permutation interval; a nonpositive or unstable contrast leaves the proposition unsupported. The test assumes the profile values are exchangeable in time within a university, which is violated under strong trends or serial dependence and is checked via the sensitivity to window length.

Scale versus structure. To describe whether the capability structure differs beyond output scale, universities are grouped into patent-volume strata and the within-stratum heterogeneity of the five-dimension profile is compared against a measurement-noise null estimated from the within-university bootstrap of each profile, rather than against zero; this separates genuine institutional differences from sampling and measurement error. The share of profile variance remaining after regression on log volume, field composition, and period is reported as descriptive evidence of variation not explained by scale or field, not as validation of the measures. The five-dimension capability structure is then compared with traditional output indicators—patent count, grant count, and forward-citation frequency (and incoPat quality ratings where available)—by partial correlation after volume control and by Kendall’s τ between the two rank orderings of universities; because no single capability score is asserted, the ranking uses a prespecified scalar composite (e.g., the first principal component of the standardized profile) reported with its uncertainty. Low partial correlation and divergent rankings are read as descriptive evidence that the two families capture different things, not as proof that one is the more valid measure.

Cases and sensitivity. Twelve trajectories sampled across profile regions and size strata are assessed by two technically qualified raters from dated patent evidence without model values; agreement, disagreements, and adjudication are reported (J. Cohen, 1960). Sensitivity covers invention unit, attribution, text representation (TF-IDF versus a frozen Chinese-capable encoder, Reimers and Gurevych, 2019), window length, citation horizon and self-citation scope, eligibility thresholds, and the E and D variants (Table 5). Common-sample agreement is reported separately from sample retention. All secondary associations are exploratory.

Results

Evidence status. The analysis below reports the capability structure and development patterns of the 124 Double-First-Class universities over 1990–2019. Numer-

A. Common technological space

Data required: IPC nodes and weighted pairs.

Position: one fixed reference layout.

Node area: fractional invention mass.

Edges: normalized co-occurrence.

Empirical map not yet computed.

B. Period overlays

Data required: period-specific IPC mass.

Reuse panel A coordinates.

Show entry, persistence, and exit.

Report retained mass and omitted nodes.

Empirical overlays not yet computed.

Figure 3. Technological knowledge space: analysis specification awaiting source data. The panels contain no synthetic nodes, invented clusters, or empirical coordinates.

Measure	Valid n	Mean	SD	Median	IQR	Zero %
Knowledge sourcing (A)	TBD	TBD	TBD	TBD	TBD	TBD
Recombination (R)	TBD	TBD	TBD	TBD	TBD	TBD
Technological exploration (E)	TBD	TBD	TBD	TBD	TBD	TBD
Recognized exploration (E^{rec} , secondary)	TBD	TBD	TBD	TBD	TBD	TBD
Knowledge diffusion (D)	TBD	TBD	TBD	TBD	TBD	TBD
Dynamic transformation (T , per transition)	TBD	TBD	TBD	TBD	TBD	—

Table 2. Distribution of the five dimensions across eligible university-periods and of T across consecutive eligible transitions. Component summaries, missing counts, and coverage shares accompany each row in the output file.

ical values are filled from the experiment outputs; entries marked [Pending:] are computed in the accompanying code and inserted at reporting time.

Coverage and landscape

The analytic population comprises 124 universities, [Pending: unique inventions], and [Pending: university-periods] in [Pending: mature cohorts]. Table 1 separates raw retrieval, deduplication, and eligibility, with the leading exclusion reasons [Pending: reasons and counts]. Of the citation layer, [Pending: share] of forward links carry citing applicant fields, which fixes the level at which b_p is computed. Figure 3 reports [Pending: node, edge, and community counts] and the retained mass; entry, persistence, and exit of technical groups are [Pending: observed patterns].

Profiles

Table 2 reports each dimension's distribution and zero share. Correlations among A , R , E , D are [Pending: coefficients and intervals], and the share of variance remaining after adjustment for volume, field, and period is [Pending: estimate]. Figure 4A shows where universities fall in profile space and which regions are populated; E^{rec} is nonzero for [Pending: share] of university-periods.

Selected component	Share	$\bar{A}$	$\bar{R}$	$\bar{E}$	$\bar{D}$	Max. post.
S_1	TBD	TBD	TBD	TBD	TBD	TBD
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$S_{\hat{K}}$	TBD	TBD	TBD	TBD	TBD	TBD

Table 3. GMM state characteristics (period-level), with the number of rows determined by model selection. Means require university-bootstrap intervals; “Max. post.” is the mean maximum posterior assignment probability. The adjusted Rand index against the a-priori patterns is computed at the university (trajectory) level after mapping each university’s state sequence to a label, on the common unit.

From / to	S_1	$\dots$	$S_{\hat{K}}$	Row count
S_1	TBD	$\dots$	TBD	TBD
$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
$S_{\hat{K}}$	TBD	$\dots$	TBD	TBD

Table 4. First-order transition matrix among the derived development-pattern labels (obtained from the GMM-state sequence under prespecified rules). Each populated probability cell must include its count and 95% university-bootstrap interval; only defined rows sum to one.

Development patterns

Among the 124 universities, the three a-priori patterns classify [Pending: counts by steady utilization, exploratory innovation, structural recombination], with [Pending: mixed] and [Pending: unsupported] cases. Cross-checking with the Gaussian mixture (Table 3) yields an adjusted Rand index of [Pending: ARI] against the a-priori assignment, indicating [Pending: agreement level]. Among [Pending: count] universities with consecutive eligible cohorts, [Pending: count] show at least one T event above its null and [Pending: count] show a sustained event; [Pending: count] events are removed by the merger and IPC-revision audit. The first-order Markov transition matrix among the derived trajectory labels (Table 4) reports [Pending: switch rates], showing how universities move between labels over time. The contrast between observed and permuted “exploration/recombination then transformation” sequences is [Pending: effect and interval]. Figure 4B displays sampled ordered profiles with their T values.

A. Profile space

Data required: A, R, E, D per eligible university-period.

Original scales; pairwise panels or a secondary PCA view for orientation. Point size: attributed invention mass. Whole-university bootstrap intervals.

No profile values computed.

B. Ordered trajectories

Data required: consecutive cohorts and T with null.

Horizontal axis: publication cohort. Rows: prespecified sampled universities. Bars: A, R, E, D ; markers: T above null, sustained events, audited mergers.

No trajectories assigned.

Figure 4. Profile and trajectory displays: analysis specification. Trajectory categories are read from the stated observable criteria and the independent T test, not from a fitted clustering.

Test family	Alternative and comparison	Observed result
Invention unit	Application versus family; common-sample profile change	TBD
Attribution	University-fractional versus all-applicant and full counts	TBD
Text representation	Character TF-IDF versus frozen Chinese-capable encoder; novelty rank agreement	TBD
Window length	Non-overlapping 2-, 3-, and 4-year cohorts; pattern-category agreement	TBD
Diffusion horizon and scope	$H = 3$ versus 5; self-citation exclusion; organization- versus family-level b_p	TBD
Eligibility	Minimum 10/20/30 inventions; completeness 70/80/90%	TBD
Exploration variant	E versus E^{rec} ; nonzero shares	TBD
Transformation null	Label permutation versus size-matched subsampling; event counts	TBD
Cases	Blinded rater agreement; cases rejecting the category reading	TBD

Table 5. Sensitivity and case results, pending computation. Each row reports the common-sample size, the change in profile values or pattern counts, an interval where applicable, and the change in coverage.

Scale versus structure

Within patent-volume strata, the five-dimension profiles show within-stratum heterogeneity that exceeds the measurement-noise null ([Pending: noise-aware p]), and [Pending: share] of profile variance remains after adjustment for log volume, field, and period. The capability structure correlates with traditional output indicators at [Pending: partial correlations], and Kendall's τ between the two university rank orderings (capability composite vs traditional composite) is [Pending: τ]; these are read as descriptive evidence that universities with similar output differ in knowledge-capability structure.

Cases and sensitivity

Rater agreement on [Pending: count] cases is [Pending: agreement and kappa]; principal disagreements concern [Pending: themes]. Table 5 reports the settings that change profile values or pattern counts most; dependence on a single representation, on sparse citations, or on one eligibility threshold bounds the interpretation.

Discussion

The framework's contribution to information science is both representational and empirical. Each inference is tied to the relation that supports it: an unusual IPC pair supports a combination claim; semantic distance from a dated reference corpus supports a technological-exploration claim; a later external citing record supports a knowledge-diffusion claim; divergence between successive category distributions, against a within-university null, supports a dynamic-transformation claim. Retaining these origins makes agreement and disagreement among representations visible and connects patent mapping (Kay et al., 2014; Leydesdorff et al., 2014; Yan & Luo, 2017) to capability measurement without collapsing the relations into one score.

Empirically, the analysis examines whether universities with similar patent output differ in their five-dimension capability structures and whether the three development patterns mark distinct paths: steady-utilization universities consolidate and diffuse an established core; exploratory-innovation universities repeatedly enter novel directions and form unusual combinations; structural-recombination universities reorganize their portfolios across periods. These paths are not a performance ranking; they describe how portfolios differ, not the causes of advantage formation, and the observational design supports differences and temporal associations rather than mechanisms.

The design also states what a university-only corpus can support. It can describe knowledge sourcing, recombination, technological exploration, and knowledge diffusion among observed citers, and it can test whether a portfolio's composition changed more than expected under period-label exchangeability within the university. It cannot establish economy-wide diffusion, reuse chains beyond one step, or the causes of change.

Other limits persist. Patents omit scientific, tacit, and unpatented knowledge. Citations do not observe learning or contractual transfer (Alcácer & Gittelman, 2006). Novelty is relative to a corpus and a representation. Institutional events such as mergers, recruitment, and policy affect both profiles and later change. Profiles should therefore be read with their coverage and uncertainty, consistent with responsible use of research metrics (Hicks et al., 2015), and not as a ranking.

Conclusion

This paper connects the contents of patent records, their dated relations, and the time at which each relation becomes observable into a framework for measuring university patent technological innovation capability from patent knowledge-association structures. Four concurrent dimensions—knowledge sourcing, recombination, technological exploration, and knowledge diffusion—are measured from separate relation layers; dynamic transformation is assessed independently as sustained compositional change. Applied to 124 Double-First-Class universities over 1990–2019, the framework is used to examine whether similar-output universities differ in capability structure and whether they fall into three development patterns—steady utilization, exploratory innovation, and structural recombination—offering a process-based lens on how innovation capability is described that stops short of causal claims about its formation. The framework is specified against the actual boundaries of a university-only patent and citation corpus, so that its empirical claims are limited to what the data observe. Extensions with enterprise citing records would allow diffusion to be traced beyond the academic system, and publication–patent links would separate scientific from technological sourcing.

Data and code availability

The source database covers invention patents and citation records for 124 Double-First-Class universities over 1990–2019; database permissions and any deposit must be verified by the authors. The accompanying package contains the manuscript, display sources, and the experiment code that produces the reported values. Provenance, output fields, and audit requirements follow traceable data stewardship principles (Wilkinson et al., 2016).

References

- Alcácer, J., & Gittelman, M. (2006). Patent citations as a measure of knowledge flows: The influence of examiner citations. *Review of Economics and Statistics*, 88(4), 774–779. <https://doi.org/10.1162/rest.88.4.774>
- Boccaletti, S., Bianconi, G., Criado, R., del Genio, C., Gómez-Gardeñes, J., Romance, M., Sendiña-Nadal, I., Wang, Z., & Zanin, M. (2014). The structure and dynamics of multilayer networks. *Physics Reports*, 544(1), 1–122. <https://doi.org/10.1016/j.physrep.2014.07.001>
- Börner, K., Chen, C., & Boyack, K. W. (2003). Visualizing knowledge domains. *Annual Review of Information Science and Technology*, 37(1), 179–255. <https://doi.org/10.1002/aris.1440370106>
- Chen, C. (2006). CiteSpace II: Detecting and visualizing emerging trends and transient patterns in scientific literature. *Journal of the American Society for Information Science and Technology*, 57(3), 359–377. <https://doi.org/10.1002/asi.20317>
- Cohen, J. (1960). A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1), 37–46. <https://doi.org/10.1177/001316446002000104>
- Cohen, W. M., & Levinthal, D. A. (1990). Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly*, 35(1), 128–152. <https://doi.org/10.2307/2393553>
- Dosi, G. (1982). Technological paradigms and technological trajectories. *Research Policy*, 11(3), 147–162. [https://doi.org/10.1016/0048-7333\(82\)90016-6](https://doi.org/10.1016/0048-7333(82)90016-6)
- Efron, B. (1979). Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1), 1–26. <https://doi.org/10.1214/aos/1176344552>
- Fleming, L. (2001). Recombinant uncertainty in technological search. *Management Science*, 47(1), 117–132. <https://doi.org/10.1287/mnsc.47.1.117.10671>
- Fraley, C., & Raftery, A. E. (2002). Model-based clustering, discriminant analysis, and density estimation. *Journal of the American Statistical Association*, 97(458), 611–631. <https://doi.org/10.1198/016214502760047131>
- Funk, R. J., & Owen-Smith, J. (2017). A dynamic network measure of technological change. *Management Science*, 63(3), 791–817. <https://doi.org/10.1287/mnsc.2015.2366>

- Griliches, Z. (1990). Patent statistics as economic indicators: A survey. *Journal of Economic Literature*, 28(4), 1661–1707. <https://www.jstor.org/stable/2727442>
- Hall, B. H., Jaffe, A., & Trajtenberg, M. (2005). Market value and patent citations. *The RAND Journal of Economics*, 36(1), 16–38. <https://www.jstor.org/stable/1593752>
- Hennig, C. (2007). Cluster-wise assessment of cluster stability. *Computational Statistics & Data Analysis*, 52(1), 258–271. <https://doi.org/10.1016/j.csda.2006.11.025>
- Hicks, D., Wouters, P., Waltman, L., de Rijcke, S., & Rafols, I. (2015). Bibliometrics: The Leiden manifesto for research metrics. *Nature*, 520(7548), 429–431. <https://doi.org/10.1038/520429a>
- Huang, Y., Bu, Y., Ding, Y., & Lu, W. (2019). From zero to one: A perspective on citing. *Journal of the Association for Information Science and Technology*, 70(10), 1098–1107. <https://doi.org/10.1002/asi.24177>
- Jaffe, A. B., Trajtenberg, M., & Henderson, R. (1993). Geographic localization of knowledge spillovers as evidenced by patent citations. *The Quarterly Journal of Economics*, 108(3), 577–598. <https://doi.org/10.2307/2118401>
- Katila, R., & Ahuja, G. (2002). Something old, something new: A longitudinal study of search behavior and new product introduction. *Academy of Management Journal*, 45(6), 1183–1194. <https://doi.org/10.2307/3069433>
- Kay, L., Newman, N., Youtie, J., Porter, A. L., & Rafols, I. (2014). Patent overlay mapping: Visualizing technological distance. *Journal of the Association for Information Science and Technology*, 65(12), 2432–2443. <https://doi.org/10.1002/asi.23146>
- Kelly, B., Papanikolaou, D., Seru, A., & Taddy, M. (2021). Measuring technological innovation over the long run. *American Economic Review: Insights*, 3(3), 303–320. <https://doi.org/10.1257/aeri.20190499>
- Kessler, M. M. (1963). Bibliographic coupling between scientific papers. *American Documentation*, 14(1), 10–25. <https://doi.org/10.1002/asi.5090140103>
- Kivela, M., Arenas, A., Barthelemy, M., Gleeson, J. P., Moreno, Y., & Porter, M. A. (2014). Multilayer networks. *Journal of Complex Networks*, 2(3), 203–271. <https://doi.org/10.1093/comnet/cnu016>

- Kogler, D. F., Rigby, D. L., & Tucker, I. (2013). Mapping knowledge space and technological relatedness in US cities. *European Planning Studies*, 21(9), 1374–1391. <https://doi.org/10.1080/09654313.2012.755832>
- Leydesdorff, L. (2008). Patent classifications as indicators of intellectual organization. *Journal of the American Society for Information Science and Technology*, 59(10), 1582–1597. <https://doi.org/10.1002/asi.20814>
- Leydesdorff, L., Kushnir, D., & Rafols, I. (2014). Interactive overlay maps for US patent (USPTO) data based on International Patent Classification (IPC). *Scientometrics*, 98(3), 1583–1599. <https://doi.org/10.1007/s11192-012-0923-2>
- Lin, J. (1991). Divergence measures based on the Shannon entropy. *IEEE Transactions on Information Theory*, 37(1), 145–151. <https://doi.org/10.1109/18.61115>
- March, J. G. (1991). Exploration and exploitation in organizational learning. *Organization Science*, 2(1), 71–87. <https://doi.org/10.1287/orsc.2.1.71>
- Nagaoka, S., Motohashi, K., & Goto, A. (2010). Patent statistics as an innovation indicator. In *Handbook of the economics of innovation* (pp. 1083–1127, Vol. 2). Elsevier. [https://doi.org/10.1016/s0169-7218\(10\)02009-5](https://doi.org/10.1016/s0169-7218(10)02009-5)
- Porter, A. L., & Rafols, I. (2009). Is science becoming more interdisciplinary? measuring and mapping six research fields over time. *Scientometrics*, 81(3), 719–745. <https://doi.org/10.1007/s11192-008-2197-2>
- Price, D. J. d. S. (1965). Networks of scientific papers. *Science*, 149(3683), 510–515. <https://doi.org/10.1126/science.149.3683.510>
- Rafols, I., Porter, A. L., & Leydesdorff, L. (2010). Science overlay maps: A new tool for research policy and library management. *Journal of the American Society for Information Science and Technology*, 61(9), 1871–1887. <https://doi.org/10.1002/asi.21368>
- Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using siamese BERT-networks. *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 3980–3990. <https://doi.org/10.18653/v1/d19-1410>
- Rigby, D. L. (2015). Technological relatedness and knowledge space: Entry and exit of US cities from patent classes. *Regional Studies*, 49(11), 1922–1937. <https://doi.org/10.1080/00343404.2013.854878>

- Rotolo, D., Hicks, D., & Martin, B. R. (2015). What is an emerging technology? *Research Policy*, 44(10), 1827–1843. <https://doi.org/10.1016/j.respol.2015.06.006>
- Sarica, S., Luo, J., & Wood, K. L. (2020). TechNet: Technology semantic network based on patent data. *Expert Systems with Applications*, 142, 112995. <https://doi.org/10.1016/j.eswa.2019.112995>
- Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27(3), 379–423. <https://doi.org/10.1002/j.1538-7305.1948.tb01338.x>
- Small, H. (1973). Co-citation in the scientific literature: A new measure of the relationship between two documents. *Journal of the American Society for Information Science*, 24(4), 265–269. <https://doi.org/10.1002/asi.4630240406>
- Small, H., Boyack, K. W., & Klavans, R. (2014). Identifying emerging topics in science and technology. *Research Policy*, 43(8), 1450–1467. <https://doi.org/10.1016/j.respol.2014.02.005>
- Stirling, A. (2007). A general framework for analysing diversity in science, technology and society. *Journal of The Royal Society Interface*, 4(15), 707–719. <https://doi.org/10.1098/rsif.2007.0213>
- Thompson, P., & Fox-Kean, M. (2005). Patent citations and the geography of knowledge spillovers: A reassessment. *American Economic Review*, 95(1), 450–460. <https://doi.org/10.1257/0002828053828509>
- Traag, V. A., Waltman, L., & van Eck, N. J. (2019). From Louvain to Leiden: Guaranteeing well-connected communities. *Scientific Reports*, 9(1), 5233. <https://doi.org/10.1038/s41598-019-41695-z>
- Uzzi, B., Mukherjee, S., Stringer, M., & Jones, B. (2013). Atypical combinations and scientific impact. *Science*, 342(6157), 468–472. <https://doi.org/10.1126/science.1240474>
- van Eck, N. J., & Waltman, L. (2010). Software survey: VOSviewer, a computer program for bibliometric mapping. *Scientometrics*, 84(2), 523–538. <https://doi.org/10.1007/s11192-009-0146-3>
- Verhoeven, D., Bakker, J., & Veugelers, R. (2016). Measuring technological novelty with patent-based indicators. *Research Policy*, 45(3), 707–723. <https://doi.org/10.1016/j.respol.2015.11.010>

- Wang, J., Veugelers, R., & Stephan, P. (2017). Bias against novelty in science: A cautionary tale for users of bibliometric indicators. *Research Policy*, 46(8), 1416–1436. <https://doi.org/10.1016/j.respol.2017.06.006>
- Whalen, R., Lungeanu, A., DeChurch, L., & Contractor, N. (2020). Patent similarity data and innovation metrics. *Journal of Empirical Legal Studies*, 17(3), 615–639. <https://doi.org/10.1111/jels.12261>
- Wilkinson, M. D., Dumontier, M., Aalbersberg, I. J., Appleton, G., Axton, M., Baak, A., Blomberg, N., Boiten, J.-W., da Silva Santos, L. B., Bourne, P. E., Bouwman, J., Brookes, A. J., Clark, T., Crosas, M., Dillo, I., Dumon, O., Edmunds, S., Evelo, C. T., Finkers, R., ... Mons, B. (2016). The FAIR guiding principles for scientific data management and stewardship. *Scientific Data*, 3(1), 160018. <https://doi.org/10.1038/sdata.2016.18>
- Yan, B., & Luo, J. (2017). Measuring technological distance for patent mapping. *Journal of the Association for Information Science and Technology*, 68(2), 423–437. <https://doi.org/10.1002/asi.23664>
- Youn, H., Strumsky, D., Bettencourt, L. M. A., & Lobo, J. (2015). Invention as a combinatorial process: Evidence from US patents. *Journal of The Royal Society Interface*, 12(106), 20150272. <https://doi.org/10.1098/rsif.2015.0272>